Dawit Kiros Redie

PhD Researcher | Machine Learning & Distributed Optimization

Trondheim, Norway | dawit.k.redie@gmail.com

Website | LinkedIn | GitHub | Google Scholar

Profile

PhD researcher at NTNU combining machine learning algorithm design, convergence analysis, and reproducible PyTorch experiments. Research focuses on communication-efficient federated learning and distributed optimization under client heterogeneity and partial participation. Applied experience spans medical image segmentation and data engineering with PySpark, Snowflake, and PostgreSQL.

Technical Skills

Programming: Python, C++, SQL, Java, Scala
ML and distributed computing: PyTorch, Flower, Ray, SLURM, GPU/HPC experiments
Data and development: Spark/PySpark, Snowflake, PostgreSQL, Spring Boot, Git, LaTeX
Methods: Federated learning, distributed optimization, compression, error feedback, experiment design

Research and Engineering Experience

Visiting PhD Researcher Aug 2026 - Present
CISPA *Saarbrücken, Germany*

- Investigate one-shot federated learning, focusing on theoretical guarantees and model performance across distributed data.

PhD Researcher Dec 2023 - Present
Norwegian University of Science and Technology (NTNU) *Trondheim, Norway*

- Design and analyze communication-efficient federated learning methods using block-wise sparsification, quantization, and error feedback.
- Study the effects of client heterogeneity and partial participation on optimization and generalization, including personalized and privacy-preserving learning.
- Develop reproducible Python/PyTorch research code and run controlled GPU/HPC sweeps to evaluate training methods.

Data Scientist Sep 2025 - Sep 2026
ReLU, in collaboration with Oslo University Hospital *Trondheim, Norway*

- Developed an abdominal aortic aneurysm segmentation pipeline for CT images using TotalSegmentator, aorta seed points, and geometric and boundary constraints.
- Developed a global controller for SO(3)-equivariant aorta tracking; contributed to dataset preparation, evaluation metrics, and reproducible ablation experiments.

Data Engineer Intern Jan 2023 - Aug 2023
Johnson Controls *Pune, India*

- Reduced data processing time by **50%** through ETL workflow optimization and migration of Spark jobs to PySpark.
- Built a Spring Boot data-loss detection service to automate manual checks and implemented data synchronization between Snowflake and PostgreSQL.
- Developed a predictive maintenance model using historical equipment data.

Selected Publications

Publications in federated learning, optimization, and medical image analysis.

1. **SA-PEF: Step-Ahead Partial Error Feedback for Efficient Federated Learning.**
   D. K. Redie, R. Arablouei, S. Werner
   Transactions on Machine Learning Research, 2026.

2. **Block-Wise Sparsification with Error Feedback for Communication-Efficient Federated Learning.**
   D. K. Redie, R. Arablouei, S. Werner
   The 34th European Signal Processing Conference (EUSIPCO), 2026.

3. **Enhancing Efficiency and Privacy in Federated Learning via Partial Model Sharing.**
   D. K. Redie, R. Arablouei, S. Werner
   59th Asilomar Conference on Signals, Systems, and Computers, 2025.

4. **Oral cancer detection using transfer learning-based framework from histopathology images.**
   D. K. Redie, S. Bilgaiyan, S. Sagnika
   Journal of Electronic Imaging, 2023.

5. **Diagnosis of COVID-19 using chest X-ray images based on modified DarkCovidNet model.**
   D. K. Redie, A. E. Sirko, T. M. Demissie, S. S. Teferi, V. K. Shrivastava, O. P. Verma.
   Evolutionary Intelligence, 2022.

Education

PhD, Statistical Signal Processing and Machine Learning at the Edge — Dec 2023 - Present
Norwegian University of Science and Technology (NTNU) — *Trondheim, Norway*

M.Tech., Computer Science Engineering — Sep 2021 - Aug 2023
Kalinga Institute of Industrial Technology (KIIT) — *Bhubaneswar, India*

B.Tech., Electronics and Computer Science Engineering — Sep 2017 - Aug 2021
Kalinga Institute of Industrial Technology (KIIT) — *Bhubaneswar, India*

Teaching Experience

Graduate Teaching Assistant, NTNU — Dec 2023 - Sep 2025
Estimation, Detection and Classification; Digital Signal Processing — *Trondheim, Norway*

- Led help sessions and individual guidance; supported assignments, grading, and examinations.

Teaching Assistant, KIIT — Sep 2022 - Jan 2023
Data Structures and Algorithms — *Bhubaneswar, India*

- Supported lectures, labs, and grading; developed C++/Python practice material on arrays, trees, graphs, and complexity.

Awards

- **Vice-Chancellor's Silver Medal** (2021): Highest CGPA in the undergraduate program, Class of 2021.
- **KIIT Merit Scholarship** (2019 - 2021): Awarded for consistently high academic performance.
- **Ethiopian Betre-Science Government Scholarship** (2017): Awarded for the highest GPA among first-year university students; supported undergraduate study abroad.